

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 05.2

Design and Analysis of an FIR High-Pass Filter Using the Hamming Window Technique in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

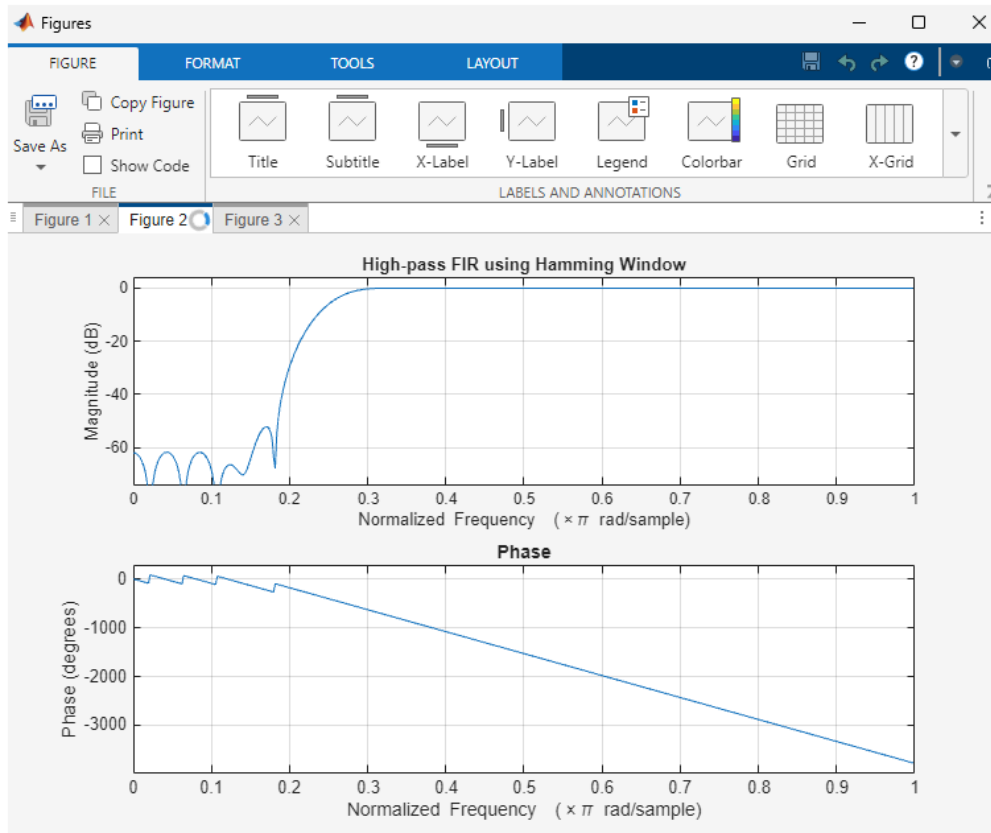
```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000;          % High-pass cutoff frequency (Hz)
fs = 8000;          % Sampling frequency (Hz)
N = 50;             % Filter order
wc = fp/(fs/2);     % Normalized cutoff (0-1)
%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);
%% --- Frequency Response Plot ---
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);
%% --- Frequency Response Plots ---
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
--- High-Pass Coefficients (Hamming Window) ---
-0.0007
-0.0000
0.0009
0.0016
0.0015
-0.0000
-0.0024
-0.0044
-0.0039
-0.0000
0.0060
0.0103
0.0088
-0.0000
-0.0128
-0.0217
-0.0184
-0.0000
0.0268
0.0464
0.0410
-0.0000
-0.0725
-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
-0.0000
0.0410
0.0464
```

0.0268
-0.0000
-0.0184
-0.0217
-0.0128
-0.0000
0.0088
0.0103
0.0060
-0.0000
-0.0039
-0.0044
-0.0024
-0.0000
0.0015
0.0016
0.0009
-0.0000
-0.0007



Result:

1. The FIR High-Pass Filter was successfully designed using the Hamming Window technique for the specified cutoff frequency, sampling frequency, and filter order.
2. The FIR filter coefficients were successfully generated using the Hamming Window method in MATLAB and the filter was implemented.
3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response showed effective attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass filtering behavior. The phase response exhibited the expected linear-phase characteristics of an FIR filter.

Practical Question 1

Design an FIR High-Pass Filter using the **Hamming Window** technique in MATLAB for a sampling frequency of **10000 Hz**, cutoff frequency of **2500 Hz**, and filter order of **50**.

Tasks:

1. Calculate the normalized cutoff frequency.
2. Generate the FIR filter coefficients using the `fir1()` function with a Hamming Window.
3. Plot the magnitude response and phase response of the designed filter.
4. Analyze the attenuation of low-frequency components and the passage of high-frequency components.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Design and implement an FIR High-Pass Filter using the **Hamming Window** technique in MATLAB for a sampling frequency of **16000 Hz**, cutoff frequency of **4000 Hz**, and filter order of **60**.

Tasks:

1. Design the FIR High-Pass Filter using the specified parameters and Hamming Window.
2. Generate and display the FIR filter coefficients.
3. Obtain the magnitude response and phase response using the `freqz()` function.
4. Evaluate the passband, stopband, and transition-band characteristics of the filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To design and analyze an FIR high-pass filter using a Hamming window and evaluate suppression of low frequencies.

Student Details

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